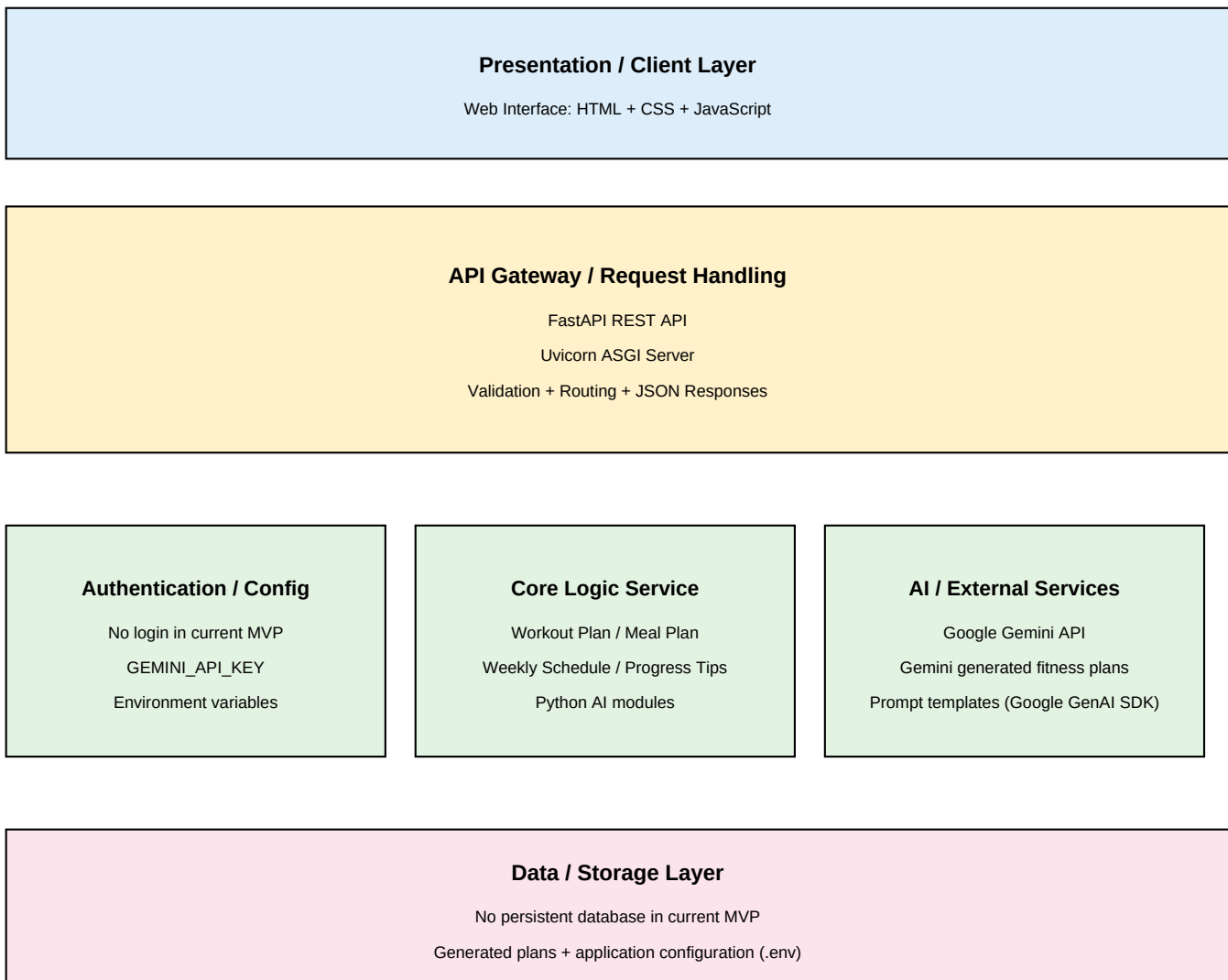


Solution Architecture

Date	29 September 2026
Team ID	SWTID-2026-3226
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies
Presentation Layer	User interface for entering fitness goals and viewing generated plans.	HTML, CSS, JS
API Gateway	Receives, validates and routes requests; returns JSON responses.	FastAPI, Uvicorn
Core Logic Service	Coordinates workout plan, meal plan, weekly schedule and progress guidance.	Python modules
AI / External Services	Generates personalized AI responses and fitness plan content.	Gemini API, Google GenAI SDK
Data / Storage	No persistent database in current MVP; generated plans and config files are used.	Local files, .env